Visible Light Communication With Event Driven Cameras

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1.1 INTRODUCTION

Visible light communication (VLC) is emerging as a low cost, low interference alternative to radio frequency (RF) communication for indoor Internet of Things (IoT) and robotics. But current VLC receivers, photodiodes and rolling shutter cameras consume significant power, struggle with fast modulation, and break down under real world lighting variations. Event driven cameras may solve these challenges, but their performance as VLC receivers remains underexplored.

Event driven cameras record only changes in brightness instead of continuously capturing full image frames. By ignoring static visual information, they react to light changes almost instantly and consume much less energy than conventional rolling shutter cameras [3]. These properties make them promising receivers for VLC, a wireless system where LEDs transmit data by rapidly modulating their brightness [1]. This research goes into how effectively event driven cameras can decode VLC signals from blinking LEDs and whether they offer advantages over traditional cameras in latency (ms), energy per bit (J/bit), and decoding accuracy measured using bit error rate (BER). By testing on-off keying (OOK) under both line-of-sight (LOS) and limited non-line-of-sight (NLOS) conditions, and by changing ambient illumination measured in lux using an external light meter, the study evaluates the potential of event based sensors as practical, low energy optical receivers [2]. Additionally, this research will explore the possibilities of color shift keying (CSK) with event driven cameras, with three potential schemes to do so. The results aim to determine whether event driven cameras can support future VLC systems for robotics, IoT devices, and neuromorphic communication technologies.

2.1 BACKGROUND

VLC enables wireless data transmission through rapid modulation of LED brightness, offering a low cost and interference free alternative to RF systems. Recent work has demonstrated that event based optical sensors can decode VLC signals by responding to rapid brightness changes rather than full image frames [2, 3, 4]. This proposal investigates whether event driven cameras, whose pixels independently report changes in brightness, can serve as efficient low power receivers for VLC.

Prior studies have shown that VLC can achieve high speed and energy efficient communication using temporal modulation (OOK) or multi-chromatic encoding schemes such as CSK [4]. The issue is that most existing VLC receivers rely on photodiodes or traditional rolling shutter cameras. These sensors need continuous frame capture or high rate sampling, resulting in high power consumption. They also suffer from motion blur, limited sampling speed, and severe degradation under uncontrolled ambient

lighting conditions [2, 6]. Event driven cameras provide an appealing alternative because they output asynchronous per pixel events with microsecond resolution, capturing only dynamic brightness changes while ignoring static information. This sparse and low latency sensing architecture suggests that event cameras may support robust, low energy VLC reception ideal for embedded and mobile systems.

Several studies demonstrate that event driven optical sensors can decode OOK based VLC by detecting abrupt brightness transitions in modulated LEDs [8, 11]. These systems report timestamped events corresponding to logical transitions and achieve reliable decoding under controlled illumination. In contrast, current literature on event camera VLC rarely evaluates multi chromatic modulation such as CSK. Traditional CMOS based CSK systems have been demonstrated using RGB LED mixtures and chromaticity mapping, achieving reduced BER compared to OOK under illumination fluctuations [10]. The only known work applying color processing to event-based sensors [9] focused on asynchronous color segmentation for vision tasks rather than modulated light communication. No prior study has examined whether event sensors can recover chromaticity information reliably enough to support CSK within a VLC system. Given that the only instance of CSK applied to event based sensors was not conducted within a VLC framework, a research gap exists regarding whether event cameras can recover chromaticity information with sufficient reliability to enable CSK in visible light communication. In addition to the lack of CSK focused studies, existing work on event-based VLC is often limited to controlled illumination conditions.

Existing event based VLC work mostly evaluates decoding under heavily controlled lighting, where ambient illumination, shadows, reflections, and NLOS effects are absent or intentionally minimized. Real indoor environments often introduce fluctuating illumination, partial occlusion, and reflective surfaces that can disrupt the detection of LED intensity changes and increase BER, yet no studies investigate how event cameras perform under these realistic conditions. Prior work rarely measures energy-per-bit (J/bit) or provides direct comparisons against rolling shutter smartphone cameras, leaving open the question of whether event cameras offer a meaningful energy advantage for VLC [8, 10].

To address these gaps, this proposal systematically evaluates event driven cameras under both controlled and realistic indoor lighting, quantifying performance using BER, latency, and energy per bit across OOK and CSK modulation schemes. By explicitly measuring decoding reliability under varying lux levels, shadows, reflections, and limited NLOS conditions, this study aims to determine whether event cameras can function as practical, robust VLC receivers that work for embedded, robotic, and neuromorphic communication systems.

2.2 TECHNICAL CONCEPT

VLC transmits data using rapid changes in LED brightness. These changes are typically encoded through modulation schemes such as OOK and CSK. In OOK, binary data are transmitted using temporal intensity changes: a high-intensity pulse represents a logical one (“1”) and the absence or low intensity pulse represents a logical zero (“0”). If the LED emits optical power $P(t)$, then the transmitted bit sequence $b(t)$ can be expressed as:

$$P(t) = \begin{cases} P_{\max}, & b(t) = 1 \\ 0, & b(t) = 0 \end{cases}$$

Alternatively, CSK transmits data through spatial color variations rather than intensity alone, mapping bits to chromaticity coordinates defined by multiple LED wavelengths. This technique supports higher data rates and greater resilience to dimming, though it requires color sensitive receivers and involves more complex decoding compared to OOK.

Unlike conventional rolling shutter cameras, event driven cameras do not capture full images or measure static intensity levels. Instead, each pixel independently reports a new event whenever the brightness change ΔL exceeds a logarithmic threshold C. Mathematically, an event is generated when:

$$|\log(L(t)) - \log(L(t - \Delta t))| > C$$

where $L(t)$ is the incident light intensity. This makes event cameras highly reactive to fast LED modulation because even subtle transitions in brightness trigger pixel level outputs with microsecond latency. The precise timestamps let the camera decode VLC signals by seeing when events happen.

Figure 1: EVK4 Event Camera schematic by Prophesee (Source: Prophesse [7])

In this work, decoding refers to the reconstruction of the transmitted digital bit sequence from the event camera's asynchronous brightness-change outputs. Rather than forming image frames or measuring absolute intensity, decoding is performed by grouping timestamped events over fixed symbol intervals and interpreting their temporal structure and relative activity. For OOK, bursts of events correspond to logical ones while low activity indicates logical zeros. The recovered bitstream is then compared to the transmitted sequence to compute bit error rate.

The quality of VLC decoding can be evaluated using BER, which measures the ratio of incorrectly decoded bits to the total transmitted bits. If N_e bits are incorrectly decoded out of N transmitted bits, the BER is expressed as:

$$\mathrm{BER} = \frac{N_e}{N}$$

3.0 EXPERIMENTAL OBJECTIVES AND EVALUATION FRAMEWORK

This section outlines a structured experimental framework to evaluate the usability of event driven cameras for embedded and optical communication systems. Because event sensors capture only brightness changes with microsecond level latency, they may offer advantages over conventional frame based CMOS cameras in decoding accuracy, energy efficiency, and resilience to ambient illumination.

Section 3.1 investigates baseline event camera timing performance and decoding reliability. Section 3.2 quantifies total energy cost per bit to determine whether event-based sensing provides measurable reductions in power consumption. Section 3.3 evaluates robustness under real-world illumination conditions. Finally, Section 3.4 isolates and extends the evaluations from 3.1 and 3.2 to VLC experiments that consist of CSK. Section 3.4 will be done in parallel to 3.1 and 3.2, as shown in figure 2, in order to allow flexibility with beginning section 3.3 with OOK experiments separately from 3.4. The CSK experiments represent a relatively unexplored area of study, and it is uncertain whether they will yield measurable results such as quantifiable bit error rates, symbol decoding accuracy, or energy-per-bit performance within the project timeline. Separating the CSK experiments into their own phase ensures that potential delays or inconclusive outcomes from this exploratory work will not impede progress on Section 3.3. Together, these studies determine whether event-driven sensors can serve as practical, low-power receivers for embedded and optical wireless communication.

In all experiments, a predefined binary sequence will be transmitted using the LED array through OOK or CSK modulation. The EVK4 event camera will record asynchronous brightness change events, which will be grouped into fixed symbol intervals to reconstruct the transmitted bitstream. The recovered sequence will then be compared to the known transmitted sequence to compute BER.

3.1 BASELINE EVENT CAMERA PERFORMANCE EVALUATION

The first objective is to characterize the intrinsic timing precision and responsiveness of the event camera under controlled optical conditions. This establishes a foundation for later VLC experiments. Key metrics include timestamp precision and latency, evaluated through controlled brightness transitions. It also includes inter-event arrival spacing and minimum detectable pulse width, determining temporal resolution limits. Finally, we will conduct event density and signal-to-noise ratio under varying illumination.

This baseline testing verifies that the event camera can capture microsecond-scale transitions

consistently and that timestamp jitter remains below the modulation symbol period. These results will define how well the camera performs on its own, before testing any specific modulation schemes.

3.2 QUANTIFYING ENERGY EFFICIENCY FOR EMBEDDED DEVICES

The second objective evaluates the total energy cost per bit (J/bit) for event-based sensing and decoding compared to frame-based approaches. Experiments will measure sensing energy (camera power consumption), compute energy (decoding and processing cost), and total energy per transmitted bit across varying data rates and illumination levels. By avoiding frame capture and reconstruction, event cameras are expected to reduce both sensing and computation energy. Demonstrating significantly lower J/bit would establish them as efficient receivers for battery-limited IoT and robotic systems, even before applying specific VLC modulation schemes.

3.3 IMPROVING ROBUSTNESS UNDER REAL-WORLD LIGHTING CONDITIONS

The third objective investigates how ambient illumination, shadows, reflections, and partial NLOS visibility affect the reliability of bit recovery from event streams. Tests will vary lux levels, measured via an external light meter, LOS and NLOS configurations, and artificial and natural lighting sources to simulate uncontrolled indoor conditions.

Performance will be quantified through BER, event timing stability, and frame drop analysis. Because event cameras respond only to brightness changes, not absolute intensity, they are expected to maintain lower BER under fluctuating illumination. If validated, this demonstrates their suitability for practical VLC in dynamic indoor environments.

3.4 VISIBLE LIGHT COMMUNICATION AND COLOR SHIFT KEYING EXPERIMENTS

The fourth objective extends the experiments from Sections 3.1 and 3.2 into a CSK context and is conducted at the end of Winter 2026 and throughout Spring 2026. These experiments run in parallel with baseline timing and energy-efficiency evaluations to allow exploratory CSK development without delaying the robustness studies in Section 3.3. While the previous sections evaluate timing precision, decoding reliability, and energy efficiency under intensity-based modulation (OOK), this section repeats those same analyses using color-encoded modulation (CSK). The goal is to determine whether event driven cameras can maintain comparable temporal accuracy, decoding performance, and energy-per-bit efficiency when processing multi-chromatic VLC signals.

LED transmitters modulate intensity or color to encode binary and multilevel data symbols. Event timestamps will be decoded to recover transmitted bit sequences. Key evaluation metrics include BER under different bitrates and illumination levels, timing accuracy and event to symbol alignment, and minimum detectable pulse width and latency variance. The goal is to keep the BER below the Forward Error Correction threshold (5–10 kbit/s), consistent with previous rolling-shutter VLC results [5, 6].

Three event-based modulation schemes will be implemented. First is spatially separated RGB LEDs (red -> right, green -> center, blue -> left). Another is frequency-encoded channels, where each color toggles at a unique frequency (e.g., red 400 Hz, green 800 Hz, blue 1200 Hz). Another is brightness-ratio encoding, mapping intensity ratios to color states (strong = red, medium = green, weak = blue). These configurations test whether event sensors can recover chromaticity information (analogous to CSK) reliably enough for multi-channel VLC. Results will compare BER, symbol timing accuracy, and energy per bit across schemes.

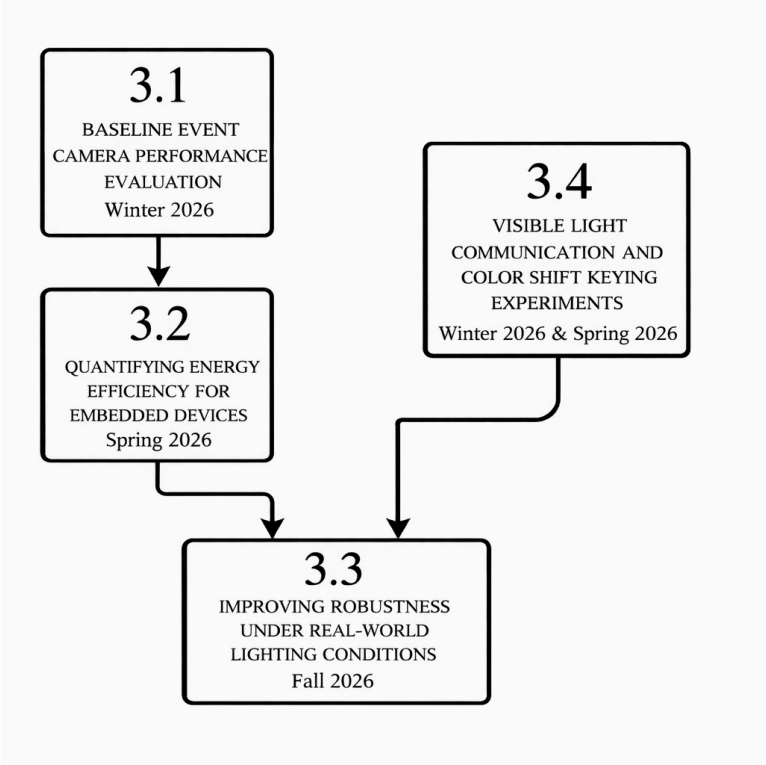

Figure 2: Flow chart demonstrating order of experimental objectives

4.1 METHODOLOGY INTRODUCTION

How will this research be conducted? This research will be conducted in two phases: controlled laboratory experiments and real world environmental evaluation. Both phases investigate VLC performance using an event driven camera as the receiver. Each stage measures decoding accuracy, latency, and energy per bit using different modulation schemes, lighting conditions, and LOS configurations. The main equipment used will consist of the EVK4 event camera, and laptop with the Prophesee Metavision software. Other equipment includes a lux meter, power meter, Google Pixel 7a, BeagleBoneBlack, and an RGB LED array.

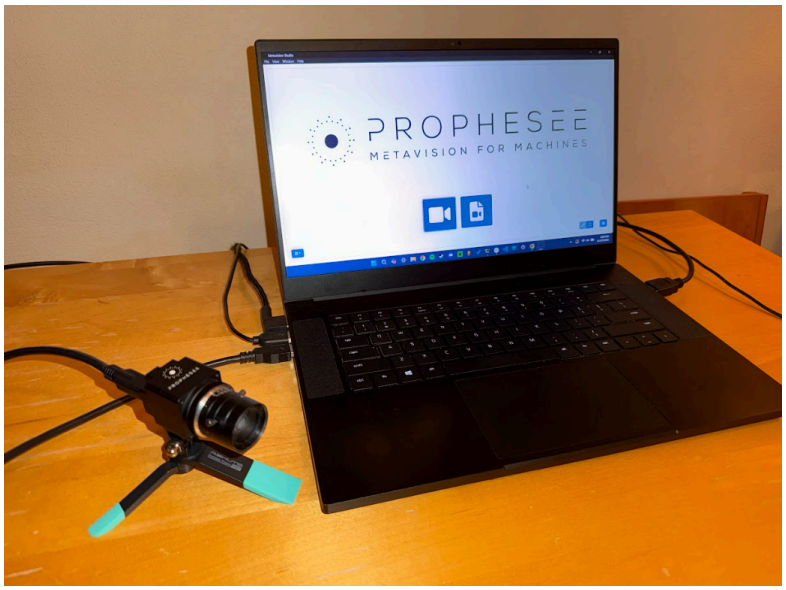

Figure 3: Setup for laboratory experiments, along with the software used

4.2 LABORATORY EXPERIMENTS

The laboratory phase will establish a controlled VLC environment to evaluate OOK and CSK under varying optical conditions. An RGB LED array driven by a programmable microcontroller, a BeagleBoneBlack, will transmit binary data sequences. Ambient illumination will be systematically adjusted using a dimmable light source, and measured with a calibrated external lux meter to quantify noise levels. The event driven camera will record asynchronous brightness changes generated by LED modulation, and the resulting data stream will be decoded into bit sequences using event timestamps. Power consumption of the camera and the decoding system will be measured using a power meter and normalized to energy per bit (J/bit). These experiments determine baseline BER, latency, and energy characteristics under known lighting thresholds.

4.3 REAL WORLD INDOOR EVALUATIONS

Upon completion of laboratory testing, the same setup will be evaluated in real world indoor environments. Testing will be performed in different rooms throughout the Westside Research Park, a hallway with mixed natural and artificial illumination, and locations that introduce partial obstruction for limited NLOS scenarios. Lighting levels will again be measured using a lux meter to correlate ambient noise and BER. The experimental objective is to quantify how lighting fluctuations, shadowing, and reflections influence decoding reliability of OOK for event based VLC receivers. Power consumption will be monitored to ensure that low energy performance persists in non controlled and random environments.

4.4 COMPARISON AGAINST CONVENTIONAL CAMERA

Finally, results from the event driven camera will be compared against those obtained using a conventional rolling shutter CMOS camera, a Google Pixel 7a, under identical lighting and modulation conditions. The comparison will include BER, latency, and energy per bit measurements, enabling an evaluation of the benefits and limitations of event driven sensors in VLC systems. This methodology

ensures that performance claims are both empirically validated and representative of real indoor communication conditions encountered by robotics, IoT sensors, and low power optical networks.

5.1 TIMELINE

Work on this project began in Fall 2025 with setup and calibration of the Prophesee EVK4 event camera. During this phase, I activated and installed the necessary software to enable working with the EVK4 camera, and established a working proficiency with the operation of the camera. Additionally, all necessary measuring and experimental tools for future experiments were collected and accounted for.

In Winter 2026, the focus shifts to full VLC decoding using the EVK4, including reconstruction of OOK and CSK signals from event streams and measurement of BER under different bitrates, LED drive conditions, and illumination levels. The objective of this phase is to demonstrate reliable bit extraction and generate BER results that can be compared to published rolling shutter CMOS benchmarks.

Spring 2026 will quantify total energy cost per bit for event based reception by measuring sensing and compute energy under matched communication settings. This produces a direct comparison between sparse event capture and frame-based processing, establishing how much energy can be saved when using an event camera as the VLC receiver.

Finally, in Fall 2026, VLC performance will be evaluated under realistic indoor lighting, including varying measured illuminance, shadows, reflections, mixed light sources, and limited NLOS. These tests determine whether event cameras maintain acceptable BER and timing fidelity in environments where conventional CMOS sensors often fail. Results from all phases will be consolidated in the senior thesis submitted in mid-December 2026.

6.1 CONCLUSION

This research proposes a low cost, energy efficient VLC system that uses event driven cameras as optical receivers. By taking advantage of their asynchronous and low power sensing architecture, this work seeks to show that event cameras can reliably decode data transmitted by modulated LEDs in both controlled and real world indoor environments. The study investigates performance under variable lighting, compares results against conventional rolling shutter cameras, and evaluates communication performance using metrics such as bit error rate, latency, and energy per bit.

The results of this project have the potential to broaden the role of event driven imaging systems further than vision only applications and into low power wireless communication. If successful, event cameras could offer a new path for optical communication in battery constrained devices, enabling smarter networks of robots, autonomous sensors in agricultural settings, and IoT systems that communicate using ambient light. By reducing power consumption while maintaining reliable data transmission, this work contributes to the development of sustainable and efficient communication technologies for future embedded and neuromorphic systems.

7.1 REFERENCES

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